

Lecture 04

Defining the Problem and Developing Alternatives

The management team of a small furniture-manufacturing company is under pressure to increase profitability to get a much-needed loan from the bank to purchase a more modern pattern-cutting machine. One proposed solution is to sell waste wood chips and shavings to a local charcoal manufacturer instead of using them to fuel space heaters for the company's office and factory areas.

- (a) Define the company's problem. Next, reformulate the problem in a variety of creative ways.
- (b) Develop at least one potential alternative for your reformulated problems in Part (a). (Don't concern yourself with feasibility at this point.)

(a) The company's problem appears to be that revenues are not sufficiently covering costs. Several reformulations can be posed:

1. The problem is to increase revenues while reducing costs.

2. The problem is to maintain revenues while reducing costs.

3. The problem is an accounting system that provides distorted cost information.

4. The problem is that the new machine is really not needed (and hence there is no need for a bank loan).

- (b) Based only on reformulation 1, an alternative is to sell wood chips and shavings as long as increased revenue exceeds extra expenses that may be required to heat the buildings. Another alternative is to discontinue the manufacture of specialty items and concentrate on standardized, high-volume products. Yet another alternative is to pool purchasing, accounting, engineering, and other white-collar support services with other small firms in the area by contracting with a local company involved in providing these services.

AN EXAMPLE PROBLEM

(Principles of engineering economy)

While studying for the Engineering Economy final exam, you and two friends find yourselves craving for a fresh pizza. You can't spare the time to pick up the pizza and must have it delivered.

- **“Pick-Up-Sticks”** offers a 1” thick (including toppings), 20” square pizza with your choice of two toppings for \$15 plus 5% sales tax and a \$1.50 delivery charge (no sales tax on delivery charge).
- **“Fred’s”** offers the round, deep-dish which is 20 inches in diameter. It is 1-3/4” thick, which includes two toppings, and costs \$17.25 plus 5% sales tax and free delivery.

- (a) What is the problem in this situation? Please state it in a clear manner.
- (b) Systematically apply the seven principles of engineering economy to the problem you have defined in part (a).
- (c) Assuming that your common unit of measure is \$ (i.e., cost), what is the better value for getting a pizza based on the criterion of minimizing cost per unit of volume?
- (d) What other criteria might be used to select which pizza to purchase?

A TYPICAL DISSCUSSION/SOLUTION

- **(a)** One problem involves how to satisfy the hunger of three students--assume a delicious pizza will satisfy this need. (Another problem is to learn enough about Engineering Economy to pass--or better yet earn an “A” or a “B”--on the final examination and top the course. Maybe a pizza will help solve this problem too?)
- Let's use “hunger satisfaction with a pizza” as the problem/need definition.

(b) Principle 1 - Develop the Alternatives

- i) **Alternative A** is to order a pizza from “Pick-Up-Sticks.”
- ii) **Alternative B** is to order a pizza from “Fred’s.”
- Other options probably exist but we’ll stick to these two alternatives.

Principle 2 - Focus on the Differences

- Difference in delivery time could be an issue.
- A perceived difference in the quality of the ingredients used to make the pizza could be another factor to consider.
- We'll concentrate our attention on cost differences in part (c) to follow.

Principle 3 - Use a Consistent Viewpoint

- Consider your problem from the perspective of three customers wanting to get a good deal. Does it make sense to buy a pizza having a crust that your dog enjoys, or ordering a pizza from a shop that employs only college students?
- Use the customers' point of view in this situation rather than that of the owner of the pizza shop or the driver of the delivery vehicle.

Principle 4 - Use a Common Unit of Measure

- Most people use “dollars” as one of the most important measures for examining differences between alternatives. In deciding which pizza to order, we’ll use a cost based metric in part (c).

Principle 5 - Consider All Relevant Criteria

- Factors other than cost may affect the decision about which pizza to order. For example, variety and quality of toppings and delivery time may be extremely important to your choice. Dynamics of group decision making may also introduce various “political” considerations into the final selection (can you name a couple?).

Principle 6 - Make Uncertainty Explicit

- The variability in quality of the pizza, its delivery time and even its price should be carefully examined in making your selection. (Advertised prices are often valid under special conditions--call first to check on this!)

Principle 7 - Revisit Your Decision

- After you've consumed your pizza and returned to studying for the final exam, were you pleased with the taste of the toppings? On the downside, was the crust like cardboard?
- You'll keep these sorts of things in mind (good and bad) when you order your next pizza!

(c) Finally some numbers to crunch--don't forget to list any key assumptions that underpin your analysis to minimize the cost per unit of volume (Principles 1, 3, 3, 4 and 6 are integral to this comparison).

Assumptions: (i) weight is directly proportional to volume (to avoid a pizza with lots of fluff but meager substance), (ii) you and your study companions will eat the entire pizza (avoids variable amounts of discarded left-overs and hence difficult-to predict cost of cubic inch consumed) and (iii) data provided in the Example Problem are accurate (the numbers have been confirmed by phone calls).

Analysis:

Alternative A “Pick-Up-Sticks”

- Volume = $20'' \times 20'' \times 1'' = 400 \text{ in}^3$
- Total Cost = $\$15 + \$0.75 + \$1.50 = \17.25
- Cost per $\text{in}^3 = \$0.043$

Alternative B “Fred’s”

- Volume = $(3.1416) (10'')^2 (1.75'') = 550 \text{ in}^3$
- Total Cost = $\$17.25 + \$0.8625 = \$18.11$
- Cost per $\text{in}^3 = \$0.033$
- Therefore, order the pizza from “Fred’s” to minimize total cost per cubic inch.

(d) Typical other criteria you and your friends could consider are:

- cost per square inch of pizza (select “Fred’s”),
- minimize total cost regardless of area or volume(select “Pick-Up-Sticks”), and
- “Fred’s” can deliver in 30 minutes but “Pick-Up-Sticks” cannot deliver for one hour because one of their ovens is not working properly (select “Fred’s”).

Example

A friend of yours bought a small apartment building for \$100,000 in a college town. She spent \$10,000 of her own money for the building and obtained a mortgage from a local bank for the remaining \$90,000. The annual mortgage payment to the bank is \$10,500. Your friend also expects that annual maintenance on the building and grounds will be \$15,000. There are four apartments (two bedrooms each) in the building that can each be rented for \$360 per month.

- (a) Does your friend have a problem? If so, what is it?
- (b) What are her alternatives? (Identify at least three.)
- (c) Estimate the economic consequences and other required data for the alternatives in Part (b).
- (d) Select a criterion for discriminating among alternatives, and use it to advise your friend on which course of action to pursue.
- (e) Attempt to analyze and compare the alternatives in view of at least one criterion in addition to cost.
- (f) What should your friend do based on the information you and she have generated?

Solution

a. Does your friend have a problem? If so, what is it?

A quick set of calculations shows that your friend does indeed have a problem. A lot more money is being spent by your friend each year ($\$10,500 + \$15,000 = \$25,500$) than is being received ($4 \times \$360 \times 12 = \$17,280$). The problem could be that the monthly rent is too low. She's losing \$8,220 per year. (Now, that's a problem!)

B- What are her alternatives? (Identify at least three.)

(b)Option (1). Raise the rent. (Will the market bear an increase?)

- Option (2). Lower maintenance expenses (but not so far as to cause safety problems).
- Option (3). Sell the apartment building. (What about a loss?)
- Option (4). Abandon the building (bad for your friend's reputation).

C- Estimate the economic consequences and other required data for the alternatives in Part (b).

- Option (1). Raise total monthly rent to $\$1,440 + \R for the four apartments to cover monthly expenses of $\$2,125$. Note that the minimum increase in rent would be $(\$2,125 - \$1,440)/4 = \$171.25$ per apartment per month (almost a 50% increase!).

- Option (2). Lower monthly expenses to \$2,125 – \$C so that these expenses are covered by the monthly revenue of \$1,440 per month. This would have to be accomplished primarily by lowering the maintenance cost. (There's not much to be done about the annual mortgage cost unless a favorable refinancing opportunity presents itself.) Monthly maintenance expenses would have to be reduced to $(\$1,440 - \$10,500/12) = \$565$. This represents more than a 50% decrease in maintenance expenses.

- Option (3). Try to sell the apartment building for \$X, which recovers the original \$10,000 investment and (ideally) recovers the \$685 per month loss ($\$8,220 \div 12$) on the venture during the time it was owned.
- Option (4). Walk away from the venture and kiss your investment good-bye. The bank would likely assume possession through foreclosure and may try to collect fees from your friend. This option would also be very bad for your friend's credit rating.

(d) Select a criterion for discriminating among alternatives, and use it to advise your friend on which course of action to pursue.

D- One criterion could be to minimize the expected loss of money. In this case, you might advise your friend to pursue Option (1) or (3).

e. Attempt to analyze and compare the alternatives in view of at least one criterion in addition to cost

e. For example, let's use "credit worthiness" as an additional criterion. Option (4) is immediately ruled out. Exercising Option (3) could also harm your friend's credit rating. Thus, Options (1) and (2) may be her only realistic and acceptable alternatives.

F- What should your friend do based on the information you and she have generated?

F- Your friend should probably do a market analysis of comparable housing in the area to see if the rent could be raised (Option 1). Maybe a fresh coat of paint and new carpeting would make the apartments more appealing to prospective renters. If so, the rent can probably be raised while keeping 100% occupancy of the four apartments.

Example

- Stan Moneymaker needs 15 gallons of gasoline to top off his automobile's gas tank. If he drives an extra eight miles (round trip) to a gas station on the outskirts of town, Stan can save \$0.10 per gallon on the price of gasoline. Suppose gasoline costs \$3.90 per gallon and Stan's car gets 25 mpg for in-town driving. Should Stan make the trip to get less expensive gasoline? Each mile that Stan drives creates one pound of carbon dioxide. Each pound of CO₂ has a cost impact of \$0.02 on the environment. What other factors (cost and otherwise) should Stan consider in his decision making?

Example

- Because each pound of CO₂ has a penalty of \$0.20,
- Savings = (15 gallons x \$0.10/gallon) - (8 lb)(\$0.020/lb) = \$1.34

- If Stan can drive his car for less than $\$1.34/8 = \0.1675 per mile, he should make the trip. The cost of gasoline only for the trip is $(8 \text{ miles} / 25 \text{ miles/gallon})(\$3.80/\text{gallon}) = \$ 1.126$,
- but other costs of driving, such as insurance, maintenance, and depreciation, may also influence Stan's decision. What is the cost of an accident, should Stan have one during his weekly trip to purchase less expensive gasoline? If Stan makes the trip weekly for a year, should this influence his decision?

Discussion